Rental-Implied Depreciation of AI Accelerators

How fast GPUs lose value, read from nine years of posted rental prices, and why the published estimates disagree.

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ABSTRACT Published estimates of how fast AI accelerators lose value run from roughly flat to about 40% a year, and hyperscalers depreciate servers over five to six years. This study rebuilds the question from 24,210 listing-months of posted rental prices for seven NVIDIA chips, V100 to B200, October 2017 to November 2026. Realized rents fall slowly: the V100's changes by −0.38 log points over nine years. The shape of the age profile depends on the market read: spot prices fall late, list prices that move fall early, and 96% of on-demand and reserved month-to-month observations do not move at all. With seven chips, valid inference leaves no strategy with a significant rate or shape. Rents stay far above operating cost, so economic life is set by obsolescence, not by rents reaching cost; a power-constrained retirement model puts it at 4.1 years.

01

Summary

Reported earnings at the largest AI spenders now depend on one estimate: how many years a server lasts. Microsoft, Alphabet, Amazon, Meta, Oracle and CoreWeave all lengthened server lives between 2020 and 2025, and Amazon partly reversed in 2025. This study asks what the rental market says about that estimate. It builds an open panel of posted GPU rents, separates age from calendar time four different ways, turns rents into asset values, and restates seven firms' earnings under the result.

24,210	96%	66%–72%	4.1 yrs
Listing-months of posted rents, 7 NVIDIA chips, Oct 2017 – Nov 2026	On-demand and reserved month-to-month observations with no price change	Coverage of the pre-registered 95% bootstrap interval in a simulation of this panel	Median retirement age in the power-constrained model; rents never fall to operating cost

KEY FINDINGS

1. **Realized rents fall slowly and unevenly.** Holding seller, region and contract fixed, the V100's rent changes by −0.13 log points over four years and −0.38 over nine. The H100's rose 0.24 in its first year, fell to −0.50 at three years and recovered to −0.04 in the 2026 shortage.
2. **Posted prices mostly measure pricing habits.** 96% of on-demand and reserved transitions show no change; AWS has charged $3.06 an hour for an on-demand V100 in every year from 2017 to 2026.
3. **The market read sets the shape.** The rate of decline in years 0–2 minus years 2–4 is 0.87 for spot prices (back-loaded) and −0.52 for list prices that change at least once (front-loaded, as the front-loading hypothesis H1 predicts).
4. **Valid inference removes the precision.** The pooled shape, 0.47, excludes zero under the pre-registered bootstrap, which a simulation shows covers only 66%–72% of the time. Under the restricted bootstrap, $p = 0.29$. No strategy that identifies a rate finds a significant decline over years 0–4.
5. **Rents never reach operating cost; book lives are not contradicted by rents.** An H100 rents for $4.50 an hour against about $0.30 of power and maintenance. Along the realized path it loses 32% of its value by year three, slower than any five- or six-year straight line.

This is a condensed edition of a 19-page working paper with a full replication package: every number in both is produced by code from archived raw files, and a fresh rebuild reproduced the paper's numbers exactly.

02

Scope and grade of evidence

Table 1. Evidence streams, their grade and what each can support.

Stream	Data	Grade	Supports
Price panel	24,210 listing-months: AWS price files, SkyPilot catalog history, archived neocloud pages, live collection from Oct 2026	Real posted prices; 94% hyperscaler; no transaction or marketplace prices	What posted rents did. Not what large buyers pay.
Age profiles	Hedonic regressions on 20,185 listing-months, four identification strategies	Real data; 7 chip clusters; specifications logged before any regression	A range of profiles under stated assumptions; no single identified rate.
Asset values	Present value of net rent on each profile, with cost assumptions	Model on real rents; hosting cost excluded	Shape of the value curve; upper bounds on value.
Triangulation	Own inference benchmark on five GPUs, reserved-price term structure, retirement Monte Carlo	Real measurements and one price snapshot; calibrated model	Economic life under power constraints.
Restatement	SEC XBRL facts and 56 quoted useful-life disclosures, 7 firms	Real filings; GPU share of capex assumed	Sign and range of earnings effects, as a grid.

03

Data

The unit is one posted price for a provider, SKU, chip, GPU count, region, contract type and term in one month, in dollars per GPU-hour. From 61,873 archived rows, zone-level spot prices become one regional daily median and daily observations one monthly median; where two sources record the same offer, the more direct one is kept. Real prices are in 2025 dollars. The estimation sample is 20,185 listing-months for seven NVIDIA chips (Table 2). Google Cloud alone is 62% of it, because it publishes every region and machine shape.

Table 2. Estimation sample by chip.

Chip	Listing-months	Providers	First month	Ages (months)	Hyperscaler / neocloud	Max / min provider
V100	5,260	6	2017-10	3–111	1.61	3.10
A100 40GB	5,510	6	2020-11	6–77	2.34	2.67
A100 80GB	3,081	6	2022-05	18–71	1.92	2.47
H100 PCIe	355	3	2023-01	4–49	2.66	1.95
H100 SXM	4,831	6	2023-01	4–49	3.10	3.52
H200	664	4	2024-12	1–23	0.73	1.73
B200	484	4	2025-06	7–23	1.54	2.08

Ages since the Epoch AI release month. The last two columns are median ratios of median on-demand prices in months where both are observed.

04

Method

For listing i of chip c in month t, log real price is modelled as

$$\ln p_{ict} = a_c + g(a_{ct}) + \delta_t + X_i'\beta + \varepsilon_{ict} \quad (1)$$

with chip effects α, month effects δ, controls for provider × region, contract × term, GPU count, price basis and form factor, and age profile g. Because age equals month minus vintage, a linear trend in age can be moved into α and δ, so the linear part of g is not identified (Hall 1968). Its curvature is, but only from differences in calendar trends across vintages. The study therefore reports each chip's realized path first, then four strategies side by side: a vintage effect priced by specifications (1a, 1b); a frontier-relative obsolescence term with no month effects (2); a successor event study (3); and a shape-only profile with the linear rate swept from 0 to −0.40 log points a year (4).

Inference is at the chip level, which leaves seven clusters. The pre-registered method was an unrestricted wild cluster bootstrap. An adversarial review, written by a reader with only the code and result tables, argued it would be too narrow; the restricted version (WCR), a test-inversion interval and a coverage simulation on this panel's design were added before writing up.

Value at age a is the present value of remaining net rent, 730 · max(u·p − o, 0) a month, discounted at r with a 2% annual failure rate, to a nine-year horizon, the oldest age observed. Operating cost o is system power × PUE × electricity plus maintenance at 4% of capital cost a year.

05

Age profiles

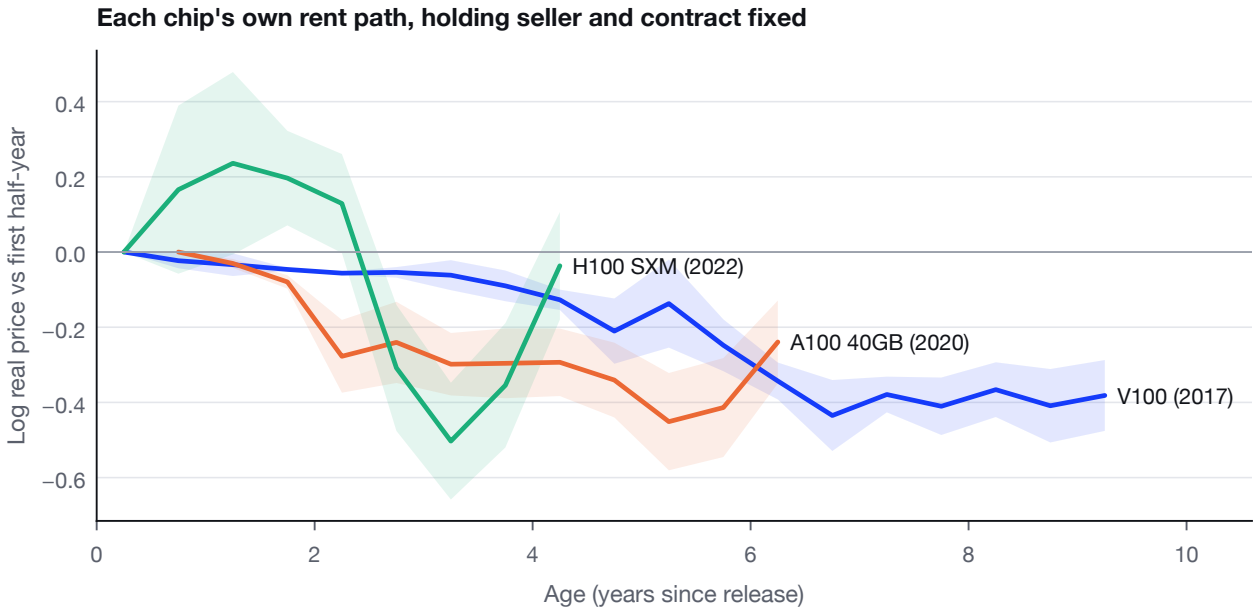


Figure 1. Observed cohort paths with 95% intervals (CR1 by provider-region), holding seller and contract fixed; no month effects.
Source: gpu-econ-life/paper/tables/p5_cohort_paths.csv, redrawn.

The realized paths need no identification assumption. They are long flat stretches, sharp falls and demand-driven rebounds, and the 2026 shortage reversed part of the decline for every chip still on sale. The projection strategies (Table 3) price a log point of TFLOP/s at 0.88 log points of rent and find no decline over years 0–2: strategy 1a's rate is 0.27 a year [−0.23, 0.77], and it turns negative (−0.11) when the H200 is left out. Every point estimate of the shape is positive.

Table 3. Age profile by strategy, log points a year. Brackets: pre-registered bootstrap; WCR p: restricted bootstrap.

Strategy and assumption	Rate, years 0–2	Rate, years 2–4	Early – late (H1: < 0)
1a · quality index (TFLOP/s) Vintage is fully priced by compute	+0.27 [−0.23, 0.77] WCR p 0.65	−0.03 [−0.24, 0.18] WCR p 0.68	+0.31 [−0.52, 1.11] WCR p 0.74
1b · three specs … by compute, bandwidth and memory	+0.28 [−0.28, 0.84] WCR p 0.59	0.00 [−0.50, 0.47] WCR p 0.99	+0.28 [−0.42, 0.97] WCR p 0.82
2 · frontier-relative Time effects work only through the best chip available	+0.11 [−0.12, 0.35] WCR p 0.49	−0.15 [−0.41, 0.11] WCR p 0.23	+0.26 [0.05, 0.47] WCR p 0.30
4 · shape only Nothing about the rate; it is swept	—	—	+0.47 [0.36, 0.57] WCR p 0.29

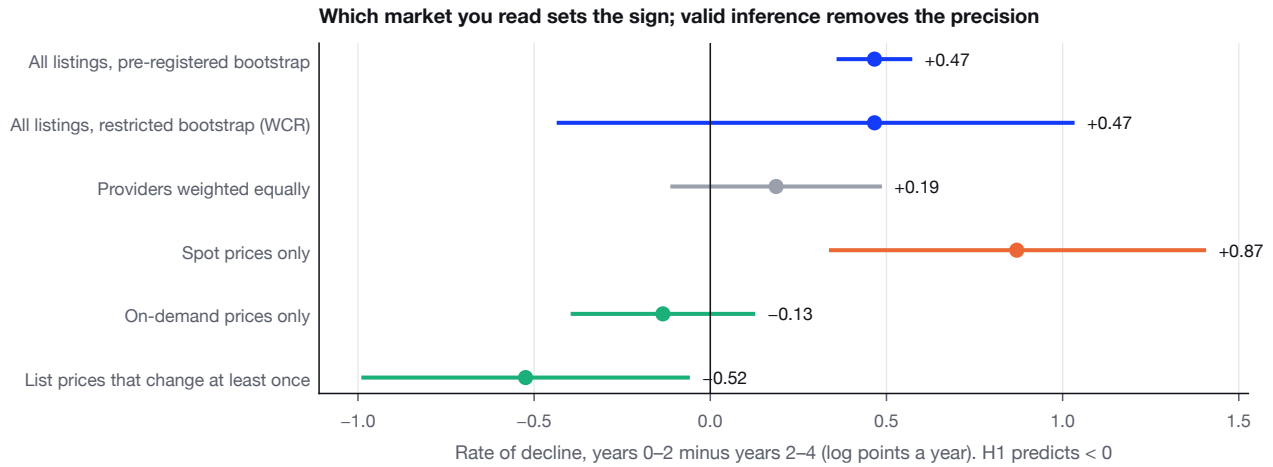


Figure 2. Strategy 4's shape by sample and inference. Intervals: pre-registered bootstrap, WCR test inversion, or ±1.96 provider-clustered standard errors (equal weights, list prices that change).

Source: gpu-econ-life/paper/tables/p5_summary.csv, p5_review_wcr_interval.csv, p5_review_weighting.csv, p5_review_repricing_shape.csv.

06

What survives valid inference

The simulation keeps this panel's rows, fixed effects and regressors and draws errors with the estimated residual structure under a true shape of zero (200 replications per design). The pre-registered intervals miss far too often; the restricted bootstrap is conservative (Table 4). Under it the pooled shape has p = 0.29 and a 95% interval of [−0.44, 1.03].

Table 4. Coverage of nominal 95% intervals for the shape when the truth is zero.

Method	Chip-month shocks i.i.d.	Chip shocks a random walk
Unrestricted wild bootstrap (pre-registered)	72.5%	66.0%
CR1, clustered by chip	76.0%	74.0%
CR1, clustered by provider	69.5%	72.5%
Restricted wild bootstrap (WCR)	98.5%	97.5%

Two further checks bound what the shape can mean. With a linear calendar trend for each chip, the quadratic term is unidentified; at a zero quadratic term the shape is -0.39. Reassigning release dates across chips, the actual assignment gives the largest shape of 1,260 assignments ($p = 0.0008$), so the curvature is tied to release order, but that test cannot separate an age effect from demand trends ordered by vintage, such as the shortages that hit the newest chips.

The successor event study finds no drop when a new generation arrives: prices in the year after a successor's release are -0.09 $[-0.40, 0.23]$ relative to the six months before, and re-dating events to the successor's first cloud listing does not change that.

THE READING

The projection strategies establish neither the speed nor the shape of the age profile in this sample. The realized paths are the better guide to the rate, and the dependence of the shape on which market is read is itself the finding that reconciles the published estimates.

07

Asset values and economic life

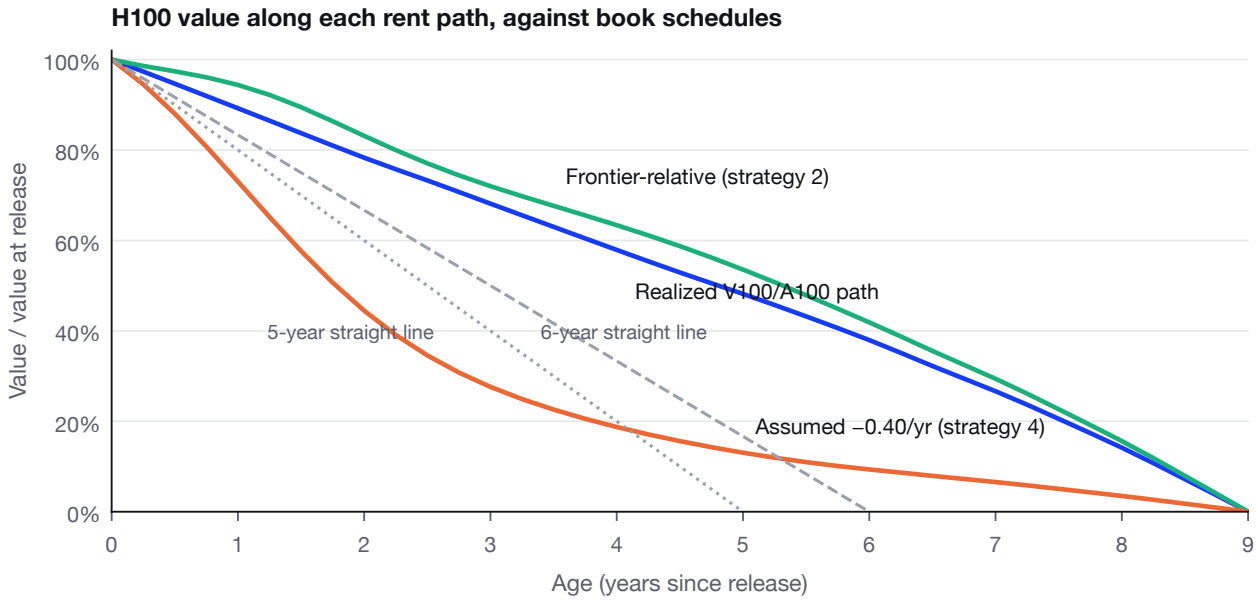


Figure 3. H100 value relative to value at release along three rent paths, base case (r 10%, utilization 70%, PUE 1.3), against straight-line schedules.

Source: gpu-econ-life/paper/tables/p6_value_curves.csv.

Table 5. H100 values by rent path, base case; range over the r × u × PUE grid in brackets.

Rent path	Implied launch rent	Value lost, year 1	By year 3	Straight-line-equivalent life (yrs)	V(0) / capex
Realized V100/A100 path	$5.85	11%	32%	9.4 (9.2–9.6)	4.5
Strategy 2, frontier-relative	$4.83	6%	28%	10.1 (9.9–10.3)	4.6
Strategy 4, assumed –0.20/yr	$6.96	15%	53%	7.1 (6.9–7.3)	5.9
Strategy 4, assumed –0.40/yr	$16.02	27%	72%	4.9 (4.8–5.0)	8.9

Straight-line-equivalent life: $2 \times \int V(a)/V(0)\, da$, the straight-line life with the same average book value. Operating cost excludes hosting, so V(0)/capex is an upper bound.

Two features matter more than the levels. First, rents never fall to operating cost within the horizon: the H100's neocloud rent of $4.50 an hour is many times its $0.30 of power and maintenance, so the end of each curve is set by the horizon. With a six-year horizon the realized path's straight-line-equivalent life is 6.1 years instead of 9.4. Rents alone do not pin down economic life; the retirement decision does. Second, only a steep assumed decline produces a front-loaded curve, and it implies an H100 launch rent of $16.02 an hour, far above any rent in the panel.

A companion analysis measures economic life three other ways (Table 6). Old chips are priced close to their measured output today. Life to operating cost is long, 9.8–11.7 years for the H100 from reserved-price ratios. In a Monte Carlo where an old chip is evicted when a newer one would earn more in the same power slot, median retirement is 4.1 years; about 47% of today's rent is scarcity markup at the calibrated level, and removing it lengthens life to 10.7 years.

Table 6. Economic life by method, years since release.

Generation (age now)	Benchmark discount vs H200	Life to opex, term structure	Retirement age, constrained (P10–P90)	Retirement, abundant power
A100 80GB (6.0)	0.00 [–0.21, 0.20]	10.7–13.2	4.1 (2.2–7.0)	11.0
H100 SXM (4.1)	–0.02 [–0.13, 0.34]	9.8–11.7	4.1 (2.2–7.0)	11.0
H200 SXM (2.0)	0.00 [0.00, 0.00]	6.5–7.5	4.1 (2.2–7.0)	11.0

08

What it means for reported earnings

No firm reports GPUs separately, so each year's cash capital expenditure from XBRL is multiplied by an assumed server share (0.6) and GPU share of server spend (0.3–0.8). For Microsoft in fiscal 2026, $115.9 billion of capital expenditure becomes a GPU base of $34.8 billion at a GPU share of one half. Each year's purchases are depreciated under the firm's disclosed lives, applied prospectively as the firms did, and under each alternative schedule.

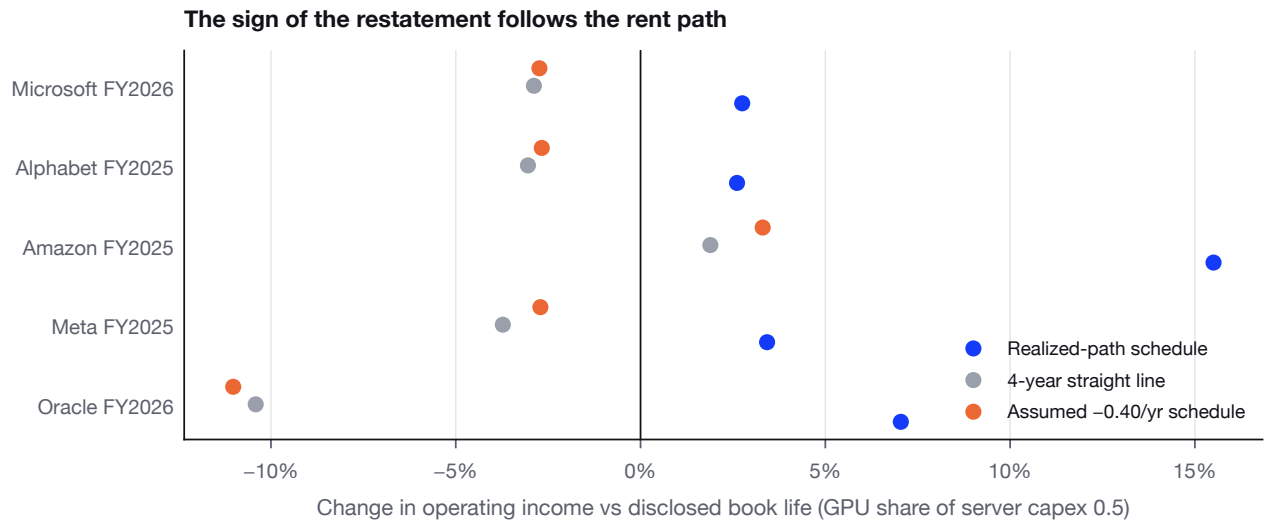


Figure 4. Change in latest-year operating income against the disclosed book life, GPU share of server capex 0.5.
Source: gpu-econ-life/paper/tables/p7_restatement.csv.

Table 7. Change in operating income, latest fiscal year (USD bn in brackets).

Firm	Realized-path schedule	4-year straight line	Assumed −0.40/yr
Microsoft FY2026	+2.8% (+4.3 bn)	−2.9% (−4.5 bn)	−2.7% (−4.2 bn)
Alphabet FY2025	+2.6% (+3.4 bn)	−3.0% (−3.9 bn)	−2.7% (−3.5 bn)
Amazon FY2025	+15.5% (+12.4 bn)	+1.9% (+1.5 bn)	+3.3% (+2.6 bn)
Meta FY2025	+3.4% (+2.9 bn)	−3.7% (−3.1 bn)	−2.7% (−2.3 bn)
Oracle FY2026	+7.0% (+1.4 bn)	−10.4% (−2.1 bn)	−11.0% (−2.3 bn)

The sign follows the rent path. Read through the realized paths, posted rents imply that book lives are, if anything, conservative: operating income would be 2.6%–15.5% higher. Only an assumed steep age decline, which the data do not establish, produces the large overstatements in the public debate; the largest is Oracle's −11.0%.

09

Robustness and limitations

- **Posted, not transacted, prices.** The panel is 94% hyperscaler list and spot prices. Marketplaces restrict automated collection and negotiated prices are private, so the results describe the posted-price market.
- **Seven clusters.** Every inference statement rests on seven chip generations; Table 4 is why the restricted bootstrap is the honest headline.
- **Weighting.** Weighting providers equally cuts the pooled shape from 0.47 to 0.19; dropping Google leaves 0.44.
- **Age zero.** Age is measured from Epoch's release month, which precedes the first cloud listing by several months for some chips.

- **Costs and shares.** Hosting, network and staff costs are excluded for lack of a sourced figure, and the restatement's server and GPU shares are assumptions shown as grids; capex excludes finance-lease additions.

10

What would settle it

A transacted-price panel by contract type, two more years of Hopper and Blackwell rents (which add vintages, the binding constraint on inference), a sourced hosting cost, and disclosure of the GPU share of server spending would each narrow the range. The panel, code and identification range are built so that each choice can be revisited as those arrive.

A

Data and reproducibility

SOURCE	gpu-econ-life: Python 3.12 with polars, DuckDB, pyfixest and statsmodels; LaTeX paper (19 pages) whose every number is a macro written from a result table.
DATA	AWS price-list offer files, SkyPilot catalog history (used, not redistributed), Wayback Machine pages for Lambda, CoreWeave and Paperspace, live collection; Epoch AI hardware data (CC BY 4.0); EIA Form 861M; BLS CPI-U; SEC EDGAR XBRL and filings. Every raw file has a sidecar with source URL, retrieval time and HTTP status, and a SHA-256 in the release manifest.
DECISIONS	26 logged methodological decisions with dates and alternatives; specifications for the age profiles logged before any regression.
VERIFICATION	529 tests; a fresh-clone rebuild from raw data reproduced every number quoted in the paper exactly, with tables matching to floating-point noise.
THIS EDITION	Figures redrawn from the committed result tables; every number here comes from those tables.

B

Definitions

EARLY − LATE RATE	Average annual log change over years 0–2 minus that over years 2–4. Negative means a front-loaded decline (H1).
WCR	Restricted wild cluster bootstrap: imposes the null hypothesis when generating draws, recommended with few, unequal clusters.
LIFE TO OPEX	Age at which rent, decaying at the implied rate, falls to operating cost.

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The code, analysis and drafts were produced with Claude (Anthropic) running in Claude Code under the author's direction; the author set the research questions and hypotheses and approved the results.